

curriculum vitae

Data Research Engineer with a Ph.D. in Mechanical Engineering and extensive experience curating large-scale datasets, building scalable Python data pipelines, and performing rigorous statistical analysis for model validation. Specialized in generating, processing, and analyzing thousands of high-fidelity simulation data points to train physics-based surrogate models. Expertise includes large-scale data ingestion, cleaning, augmentation, and ablation studies, along with uncertainty quantification, interactive visualization tools, and scalable workflow automation using Python in high-performance computing environments. Proven ability to translate complex technical findings into actionable insights for multidisciplinary teams.

Personalia

Surname: Ruybalid
Given names: Andre, Paul
Degrees: Ph.D. | MSc. | BSc.
City: Los Alamos, New Mexico, U.S.A.
Phone number: +1 719 210 1746 (personal mobile)
E-mail: andreruybalid@gmail.com
Date of birth: 26 October 1987
Place of Birth: Farmington, New Mexico, U.S.A.
Nationality: American/Dutch

Professional experience

2023 - Present

R&D Engineer, Los Alamos National Laboratory

Materials Science and Technology Division (MST-8) | Los Alamos, NM, U.S.A.

- ❖ Designed and executed large-scale simulation pipelines generating thousands of low-length-scale simulations, curating datasets for training data-driven surrogate models.
- ❖ Developed custom Python tools for data extraction, cleaning, analysis, and visualization to support model validation and decision-making.
- ❖ Performed statistical analysis and large-scale uncertainty quantification (UQ) to assess model sensitivity and improve prediction reliability.
- ❖ Built and maintained data processing workflows for nuclear fuel performance analysis, ensuring high-quality inputs for modeling and simulation.
- ❖ Collaborated with mechanistic modelers and external partners to deliver pragmatic data-driven solutions; led technical reports and peer-reviewed publications.

2021 – 2023

Postdoctoral Researcher, Los Alamos National Laboratory

Materials Science and Technology Division (MST-8) | Los Alamos, NM, U.S.A.

- ❖ Curated and processed large datasets from thousands of physics-based simulations to train surrogate models for viscoplastic material behavior.
- ❖ Implemented data ablation experiments and validation frameworks to assess dataset quality and model performance.

2019 – 2020

Design Engineer, ASML

DUV Def & Metrology group of Defectivity Department | Eindhoven, The Netherlands

Developed and maintained complex defect metrology algorithms in custom MATLAB tooling for high-precision nanoscale defect detection on DUV lithography reticles. Designed, validated, and optimized numerical algorithms through cleanroom experimentation and large-scale data analysis, delivering results under tight deadlines in collaboration with cross-functional and industrial teams.

2013 – 2019

PhD Researcher, Eindhoven University of Technology

Computational and Experimental Mechanics | Prof. Dr. Marc Geers & Dr. Johan Hoefnagels.

Philips Research, Materials Innovation Institute | Eindhoven, The Netherlands

- ❖ Developed integrated experimental-numerical frameworks combining finite element modeling, image processing, and digital image correlation for full-field data analysis.

- ❖ Created custom data processing and inverse optimization algorithms for identifying constitutive parameters from complex experimental datasets.

2017 – 2020

Account manager & Technical recruiter

Research-Square & EuFlex | Eindhoven, The Netherlands

Managed client accounts and conducted technical recruitment for highly specialized PhD-level engineering and scientific roles. Developed and implemented Python-based data analysis tools and workflows to streamline candidate sourcing, evaluation, and tracking processes.

2018 – 2019

Lab technologist, Multi-Scale Laboratory

Eindhoven University of Technology | Eindhoven, The Netherlands

Computational and Experimental Mechanics

Managed an experimental mechanics laboratory, overseeing high-tech equipment including scanning electron microscopes and micro-mechanical test systems, while providing technical guidance to researchers and students and supporting experimental design in a high-demand environment.

Education

Eindhoven University of Technology, the Netherlands

Department of Mechanical Engineering

Division: Computational and Experimental Mechanics

Research Group: Mechanics of Materials

2013 – 2019

PhD. in Mechanical Engineering

Title: "Identification of multi-layer interface systems: a full-field, micro-mechanical approach."

2013 – 2019

Graduate School of Engineering Mechanics

Advanced Mechanics Courses

2010 – 2013

MSc. Mechanical Engineering

Minor: Biomedical Engineering

2006 – 2010

BSc. Mechanical Engineering

2000 – 2006

Research-oriented secondary education (VWO)

St.-Janscollege, Hoensbroek, The Netherlands

Selected journal publications (peer-reviewed)

P. Robbe, A.P. Ruybalid, A. Hegde, C. Bonneville, H.N. Najm, L. Capolungo and C. Safta . A comparison of surrogate constitutive models for viscoplastic creep simulation of HT-9 steel. *Computational Materials Science*, 262, 114367 (2026). <https://doi.org/10.1016/j.commatsci.2025.114367>.

A.P. Ruybalid, A. Tallman, W. Wen, C. Matthews, L. Capolungo. Data-Driven Surrogate Modeling with Microstructure-Sensitivity of Viscoplastic Creep in Grade 91 Steel. Integrating Materials and Manufacturing Innovation 13, 895–914 (2024). <https://doi.org/10.1007/s40192-024-00377-z>

A.P. Ruybalid, O. van der Sluis, M.G.D. Geers and J.P.M. Hoefnagels. Full-field identification of mixed-mode adhesion properties in a flexible, multi-layer microelectronic material system. *Engineering Fracture Mechanics*, 226, pp. 106879 (2020), <https://doi.org/10.1016/j.engfracmech.2020.106879>.

A.P. Ruybalid, O. van der Sluis, M.G.D. Geers, J.P.M. Hoefnagels. An *In-Situ*, Micro-Mechanical Setup with Accurate, Tri-Axial, Piezoelectric Force Sensing and Positioning. *Exp Mech* 60, 713–725 (2020), <https://doi.org/10.1007/s11340-020-00583-8>.

A.P. Ruybalid, J.P.M. Hoefnagels, O. van der Sluis, M.P.F.H.L. van Maris and M.G.D. Geers. Mixed-mode cohesive zone parameters from integrated digital image correlation on micrographs only. *International Journal of Solids and Structures*, 156-157, pp. 179-193 (2019), <https://doi.org/10.1016/j.ijsolstr.2018.08.010>.